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Proof. Let $\alpha = \mathrm{rad}(\mathfrak{a})$. Then, for any $a \in \mathfrak{a}$, b is algebraic over (a, α) and hence, by Lemma 2.13, is algebraic over a. Since there is an $a \in \mathfrak{a}$ such that $a_i \notin \mathrm{acl}(a_{\tilde{i}}, b)$ for all $i \in [n]$, by Lemma 2.14, b must be algebraic. □

Lemma 2.19 ([16, Lemma 4.15])**.** *If $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated then every algebraic (respectively, definable) closed disc contains an algebraic (respectively, definable) point.*

Lemma 2.20 ([16, Lemma 4.17])**.** *If $\mathbb{S}$ is (VF, Γ)-generated then every algebraic (respectively, definable) disc contains an algebraic (respectively, definable) point.*

A pillar of the structure of definable sets in $\mathbb{U}$ is *C-minimality*, meaning that every definable subset of VF is a finite boolean combination of valuative discs, allowing additional parameters to name the individual discs.

Resolved referee R027 [Clarification] – Parameters of the balls in C-minimality.
Original source line 859. The balls in a finite Boolean decomposition need not be individually $\mathbb{S}$-definable. For example, a definable pair of conjugate algebraic points need not split into two definable singleton balls. Say 'balls with additional parameters (whose finite canonical family is algebraic over the defining parameters)' and retain the Galois-invariant family where individual components cannot be chosen.

Author resolution. Unqualified definability remains relative to the fixed ground parameter set. The individual discs in a finite Boolean decomposition may require additional parameters to be named; the sentence now makes that distinction explicit.

Corollary 2.21. *Suppose that $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated. Let $\mathfrak{a} \subseteq \mathrm{VF}$ be a definable disc and A a definable subset of* VF*. If $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable point.*

Proof. It is not hard to see that, by C-minimality, if $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable closed disc and hence the claim is immediate by Lemma 2.19. □

Definition 2.22. Let D be a set of parameters. We say that a (not necessarily definable) nonempty set A *generates a (complete) D-type* if, for every D-definable set B, either $A \subseteq B$ or $A \cap B = \emptyset$. In that case, A is *D-type-definable* if no set properly contains A and also generates a D-type. If A is D-definable and generates a D-type, or equivalently, if A is both D-definable and D-type-definable then we say that A is *D-atomic* or *atomic over D*.

We simply say "atomic" when $D = \emptyset$.

In the literature, a type could be a partial type and hence a type-definable set may have nontrivial intersection with a definable set. In this paper, since partial types do not play a role, we shall not carry the superfluous qualifier "complete" in our terminology.

Lemma 2.23 ([18, Lemma 3.3])**.** *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let A be an atomic set. Then A is γ-atomic for all $\gamma \in \Gamma$.*

Resolved referee R028 [Error] – Atomic base change needs a nontrivial base value group.
Original source line 879. The unrestricted statement remains false over a trivially valued base, as the human referee observed. Over a trivially valued algebraically closed field, $A = \mathcal{M} \setminus \{0\}$ is atomic; a parameter $\gamma > 0$ cuts it into the nonempty proper subset $\{x : 0 < \mathrm{val}(x) < \gamma\}$. Impose a nontrivial base value group here and propagate that assumption through uses of this lemma. In the cited proof, the essential step is that a nonempty definable subset of Γ contains a definable point; this fails for $(0, \infty)$ over the trivial base.

Author resolution. The author adopts nontrivial $\Gamma(\mathbb{S})$ as a standing assumption throughout the paper, matching the blanket hypothesis in [18]. It persists under parameter extension and supplies the

required hypothesis here and in subsequent uses, including Lemma 2.25 and Corollary 2.30. The contrary permission at the opening of Section 6 is removed.

Lemma 2.24. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc. Then $\mathfrak{a}$ is t-atomic for all $t \in \mathrm{RV}$.*

Proof. For the case $n = 1$, if $\mathfrak{a}$ were not t-atomic then, by Corollary 2.21 and Lemma 2.13, it would contain a point algebraic over the fixed ground base. This point belongs to a finite ground-definable set. Atomicity would then force the whole infinite open disc $\mathfrak{a}$ to lie in that finite set, a contradiction. For the case $n > 1$, by induction on n, $\mathfrak{a}_1$ is t-atomic and for every $a \in \mathfrak{a}_1$, $\mathfrak{a}_{>1}$ is (a, t)-atomic. So $\mathfrak{a}$ cannot have a nonempty proper t-definable subset. □

Partly resolved referee R029 [Gap] – Atomicity proof: algebraic versus definable.
Original source line 886. Lemma 2.13 only gives an algebraic point over the original base, not necessarily a definable point. This suffices: an infinite atomic disc cannot meet a finite base-definable set. Replace 'definable point' at this step by that argument. The final sentence also needs 'nonempty proper t-definable subset'. Finally, indicate why Corollary 2.21 remains applicable whenever previously named disc parameters occur in the base; its stated hypothesis excludes such extra parameters.

Author resolution. The author approves both proof corrections: the point is algebraic over the fixed ground base, so atomicity would place the infinite disc inside a finite ground-definable set; and the final sentence now says "nonempty proper." The statement is unchanged. The separate question about applicability over additional disc-imaginary ground parameters remains open.

Lemma 2.25. *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let $A \subseteq \mathrm{VF}^n$ be a set that generates a type and $\mathfrak{b} \subseteq A$ an open (or closed) polydisc. Then $\mathfrak{b}$ is $\ulcorner \mathfrak{b} \urcorner$-atomic.*

Proof. We do induction on n. The base case $n = 1$ is just [18, Lemma 3.4]. For the case $n > 1$, write $\mathfrak{b} = \prod_i \mathfrak{b}_i$, $\mathfrak{b}' = \prod_{i>1} \mathfrak{b}_i$, and $\gamma = \mathrm{rad}(\mathfrak{b}_1)$. For any $a \in \mathfrak{b}_1$, A_a generates a type. So, by Lemma 2.23 and the inductive hypothesis, $\mathfrak{b}'$ is $(a, \gamma, \ulcorner \mathfrak{b}' \urcorner)$-atomic and hence $\ulcorner \mathfrak{b} \urcorner$-atomic. Similarly $\mathfrak{b}_1$ is $\ulcorner \mathfrak{b} \urcorner$-atomic. It follows that $\mathfrak{b}$ must be $\ulcorner \mathfrak{b} \urcorner$-atomic. □

Lemma 2.26. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc and $f : \mathfrak{a} \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function such that $\mathrm{rv} \restriction f(a)$ is injective for every $a \in \mathfrak{a}$. Then $\mathrm{rv} \circ f$ is constant.*

Proof. This is clear since, by Lemma 2.24, for every $t \in \mathrm{rv}(\bigcup f(\mathfrak{a}))$, $\mathfrak{a}$ is t-atomic and hence f induces a t-definable function $\mathfrak{a} \longrightarrow t^\sharp$. The lemma follows. □

A *thin annulus* is a set of the form $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, where $a \in \mathrm{VF}$ and $\gamma \in \Gamma$. Equivalently, it is a closed disc with one maximal open subdisc removed.

Lemma 2.27 ([18, Lemma 3.8])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc or an atomic closed disc or an atomic thin annulus. Let $f : A \longrightarrow \mathrm{VF}$ be a definable function. Then $f(A)$ is also atomic of one of these three forms.*

Resolved referee R163 [Clarification] – Define thin annuli before using them.
Original source line 907. The term "thin annulus" has not been defined in this manuscript. Specify a closed disc with one maximal open subdisc removed; after choosing the center of the removed subdisc, it is a shell $\{x : \mathrm{val}(x - a) = \gamma\}$. This makes the alternatives here and the later component argument unambiguous.

Author resolution. The author approves the finite-radius shell definition $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, with $\gamma \in \Gamma$. It is now stated before the first use in Lemma 2.27.

Lemma 2.28 ([18, Lemma 3.13])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc and $f : A \longrightarrow \mathrm{VF}$ a definable function. Then either f is constant or $f(A)$ is also an atomic open disc.*

Lemma 2.29. *Let $\mathfrak{a}$, f be as in Lemma 2.26 and suppose that f is not constant. Then there are $t_1, \ldots, t_k \in \mathrm{RV}$ and for each i, a t_i-definable function $f_i : \mathfrak{a} \longrightarrow t_i^\sharp$ such that $f = \bigcup_i f_i$ and each $f_i(\mathfrak{a})$ is either a point or a t_i-atomic open disc.*

Proof. By Lemmas 2.24 and 2.26, this is immediately reduced to the case that f is a definable nonconstant function $\mathfrak{a} \longrightarrow t^\sharp$. We do induction on n. The base case $n = 1$ is just Lemma 2.28. For the case $n > 1$, by Lemma 2.27, $f(\mathfrak{a})$ is an open disc or a closed disc or a thin annulus. Suppose for contradiction that $f(\mathfrak{a})$ is a closed disc or a thin annulus. By the inductive hypothesis, for every $a \in \mathrm{pr}_1(\mathfrak{a})$ there is a maximal open subdisc $\mathfrak{b}_a \subseteq f(\mathfrak{a})$ that contains $f(\mathfrak{a}_a)$, similarly for every $a \in \mathrm{pr}_{>1}(\mathfrak{a})$. It follows that $f(\mathfrak{a})$ is actually contained in a maximal open subdisc of $f(\mathfrak{a})$, which is absurd. □

Referee R030 [Gap; revised] – A unary image lemma is used on a polydisc.
Original source line 918. For $n > 1$, Lemma 2.27 does not directly apply to $f : \mathfrak{a} \longrightarrow \mathrm{VF}$, since it has a unary domain. Supply the missing argument that $f(\mathfrak{a})$ has just one primitive component. A possible route is the canonical finite component decomposition of an atomic unary set: every coordinate slice of $\mathfrak{a}$ is atomic over its fixed coordinates, and its image is a point or an open disc by induction. Show that such an image must lie in one primitive component, then connect any two source points by changing coordinates one at a time. The same coordinate-chain argument excludes a closed disc or thin annulus as the whole image, because each slice image lies in a single maximal open subdisc. This fills the role of the inapplicable unary citation, once the component claim is spelled out.

Corollary 2.30. *Let $A \subseteq \mathrm{VF}^n$ be a definable set and $f : A \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function as in Lemma 2.26. Then there are a definable finite partition $(A_i)_i$ of A and, for each i and all open polydiscs $\mathfrak{a} \subseteq A_i$, finitely many $t_{\mathfrak{a}j}$-definable functions $f_{\mathfrak{a}j} : \mathfrak{a} \longrightarrow t_{\mathfrak{a}j}^\sharp$, where $t_{\mathfrak{a}j} \in \mathrm{RV}$, such that $f \restriction \mathfrak{a} = \bigcup_j f_{\mathfrak{a}j}$ and each $f_{\mathfrak{a}j}(\mathfrak{a})$ is either a point or an open disc.*

Proof. For $a \in A$, let $D_a \subseteq A$ be the type-definable subset containing a. By Lemma 2.25, every open polydisc $\mathfrak{a} \subseteq D_a$ is $\ulcorner \mathfrak{a} \urcorner$-atomic and hence, by Lemma 2.28, the assertion holds for $\mathfrak{a}$. Then, by compactness, the assertion must hold in a definable subset $A_a \subseteq A$ that contains a; by compactness again, it holds in finitely many definable subsets $A_1, \ldots, A_m$ of A with $\bigcup_i A_i = A$. Then the partition of A generated by $A_1, \ldots, A_m$ is as desired. □

Referee R031 [Error] – The local branch functions need the disc parameter.
Original source line 925. The functions $f_{\mathfrak{a}j}$ must be allowed parameters $(\ulcorner \mathfrak{a} \urcorner, t_{\mathfrak{a}j})$, not just $t_{\mathfrak{a}j}$. Their domains are $\mathfrak{a}$, so literal $t_{\mathfrak{a}j}$-definability would imply that every such disc is $t_{\mathfrak{a}j}$-definable. Already for $f(a) = \{1\}$ on $A = \mathrm{VF}$, this is impossible for arbitrary subdiscs, since $t_{\mathfrak{a}j} = \mathrm{rv}(1)$ is fixed. The proof itself introduces $\ulcorner \mathfrak{a} \urcorner$, which explains the missing parameter. Also replace the appeal to the unary Lemma 2.28 by the corrected multidimensional Lemma 2.29, and explicitly formulate the uniform first-order property used in the compactness step.

Definition 2.31. Let $f : A \longrightarrow B$ be a bijection between two sets A and B, each with exactly one VF-coordinate. We say that f is *concentric* if, for every VF-fiber f_t of f and all open polydiscs $\mathfrak{a} \subseteq \mathrm{dom}(f_t)$, $f_t(\mathfrak{a})$ is also an open polydisc; if both f and f^{-1} are concentric then f has the *disc-to-disc property*.

Lemma 2.32 ([18, Proposition 3.19])**.** *Let $f : A \longrightarrow B$ be a definable bijection between two sets A and B, each with exactly one* VF*-coordinate. Then there is a definable finite partition $(A_i)_i$ of A such that every restriction $f \restriction A_i$ has the disc-to-disc property.*

2.3. **Continuity and fiberwise properties.**

Definition 2.33. We set $\dim_{\mathrm{RV}}(\varnothing) = -\infty$. For a nonempty definable subset $U \subseteq \mathrm{RV}_\infty^m$, the RV-*dimension*, denoted by $\dim_{\mathrm{RV}}(U)$, is the smallest number k such that there is a definable finite-to-one function $f : U \longrightarrow \mathrm{RV}^k$ (RV^0 is taken to be the singleton $\{\infty\}$).

Resolved referee R164 [Clarification; new] – Dimension of the empty set.
Original source line 941. Specify $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$ (or another negative convention) separately. The VF definition otherwise has no largest natural number for the empty set, whereas the literal RV definition gives it dimension 0. With that latter convention the boundary lemma fails when $n = 0$, since its boundary is empty but $0 < 0$ is false. Alternatively restrict that lemma to $n \geq 1$, while still specifying the empty VF dimension.

Author resolution. The author approves $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$. Both definitions now state this convention separately and retain their original clauses for nonempty sets, including zero-coordinate cases.

Definition 2.34. Let A be a subset of VF^n. The RV-*boundary* of A, denoted by $\partial_{\mathrm{RV}} A$, is the definable subset of $\mathrm{rv}(A)$ such that $t \in \partial_{\mathrm{RV}} A$ if and only if $t^\sharp \cap A$ is a proper nonempty subset of $t^\sharp$. The definable set $\mathrm{rv}(A) \smallsetminus \partial_{\mathrm{RV}} A$, denoted by $\mathrm{int}_{\mathrm{RV}}(A)$, is called the RV-*interior* of A.

Referee R032 [Typo] – Definability of the RV-boundary.
Original source line 945. Insert 'definable' before 'subset of VF^n'; without that assumption the boundary and interior need not be definable.

Obviously, $A \subseteq \mathrm{VF}^n$ is an RV-pullback if and only if $\partial_{\mathrm{RV}} A$ is empty.

Lemma 2.35. *Let A be a definable subset of VF^n. Then $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}} A) < n$.*

Proof. We proceed by induction on n. The base case $n = 1$ follows immediately from C-minimality since $t^\sharp$ contains a t-definable proper subdisc for every $t \in \partial_{\mathrm{RV}} A$.

For the inductive step, since $\partial_{\mathrm{RV}} A_a$ is finite for every $i \in [n]$ and every $a \in \mathrm{pr}_{\widehat{i}}(A)$, by Corollary 2.21 and compactness, there are a definable finite partition $(A_{ij})_j$ of $\mathrm{pr}_{\widehat{i}}(A)$ and, for each A_{ij}, a definable finitary function $f_{ij} : A_{ij} \longrightarrow \mathcal{P}(\mathrm{VF})$ such that for all $a \in A_{ij}$, $\mathrm{rv}(f_{ij}(a)) = \partial_{\mathrm{RV}} A_a$ if A_a is not an RV-pullback or $f_{ij}(a) = \{0\}$ otherwise. By Corollary 2.30, we may assume that if $t^\sharp \subseteq A_{ij}$ then the restriction $\mathrm{rv} \upharpoonright f_{ij}(t^\sharp)$ is constant. Hence each f_{ij} induces a definable finitary function $C_{ij} : \mathrm{int}_{\mathrm{RV}}(A_{ij}) \longrightarrow \mathcal{P}(\mathrm{RV}_\infty)$. Let

$$C = \bigcup_{i,j} C_{ij} \quad \text{and} \quad B = \bigcup_{i,j} \bigcup_{t \in \partial_{\mathrm{RV}} A_{ij}} \mathrm{rv}(A)_t.$$

Obviously $\dim_{\mathrm{RV}}(C) < n$. By the inductive hypothesis, for all A_{ij} we have $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}} A_{ij}) < n-1$. Thus $\dim_{\mathrm{RV}}(B) < n$. Since $\partial_{\mathrm{RV}} A \subseteq B \cup C$, the claim follows. □

Referee R033 [Gap] – RV-boundary proof needs uniform branches.
Original source line 962. When constructing f_{ij}, ensure that each fiber contains exactly one representative of each boundary RV-disc; injectivity of rv on these finite fibers is a hypothesis of Corollary 2.30. The sentence about constancy should mean that the finite set $\mathrm{rv}(f_{ij}(a))$ is independent of $a \in t^\sharp$, not that rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet A partially; state the dimension/type fact precisely if using this route.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

Corollary 5.24. *In $\tilde{\mathbb{C}}$ with $\mathbb{S} = \mathbb{C}(\!(t)\!)$, if f is a morphism of varieties over $\mathbb{C}$ with $\mathrm{Vol}_b([A_b]) = 0$ for all $b \in B$ then $\mathrm{Vol}([A]) = 0$.*

6. Special covariant bijections

From here on, we also assume that the substructure $\mathbb{S}$ of parameters is VF-generated. So every definable disc contains a definable point (Lemma 2.20). Another simple fact we shall occasionally use is that if $\gamma \in \Gamma$ is definable then there is a $\gamma' \geq \gamma$ such that $\gamma'^{\sharp}$ contains a definable point.

6.1. Invariance and covariance.

Definition 6.1. For $\gamma \in \Gamma$, let $\pi_\gamma : \mathrm{VF} \longrightarrow \mathrm{VF}/\mathcal{M}_\gamma$ be the quotient map. If $\gamma = (\gamma_1, \ldots, \gamma_n) \in \Gamma^n$ then π_γ denotes the product of the maps π_{γ_i}.

Let $\alpha \in \Gamma^n$ and $\beta \in \Gamma^m$. We say that a function $f : A \longrightarrow B$ with $A \subseteq \mathrm{VF}^n$ and $B \subseteq \mathrm{VF}^m$ is *(α, β)-covariant* if it (π_α, π_β)-contracts to a function $f_\downarrow : \pi_\alpha(A) \longrightarrow \pi_\beta(B)$, that is, $\pi_\beta \circ f = f_\downarrow \circ \pi_\alpha$. For simplicity, we shall often suppress parameters in notation and refer to (α, β)-covariant functions as $(\alpha, -)$-covariant or $(-, \beta)$-covariant or just covariant functions.

A set in VF is *α-invariant* if its characteristic function is $(\alpha, 0)$-covariant.

More generally, for sets A, B with RV-coordinates, a function $f : A \longrightarrow B$ is *covariant* if every one of its VF-fibers is covariant (recall Terminology 2.6). Similarly, a set is *invariant* if every one of its VF-fibers is invariant.

Definition 6.2. A definable function $f : A \longrightarrow B$ is *proper covariant* if

- the sets A_{VF}, $f(A)_{\mathrm{VF}}$ are bounded and the sets A_{RV}, $f(A)_{\mathrm{RV}}$ are doubly bounded,
- for each VF-fiber f_t of f there is a t-definable tuple $(\alpha_t, \beta_t) \in \Gamma$ such that f_t is (α_t, β_t)-covariant, $\mathrm{dom}(f_t)$ is α_t-invariant, and $\mathrm{ran}(f_t)$ is β_t-invariant.

A definable set A is *proper invariant* if A_{VF} is bounded, A_{RV} is doubly bounded, and for each VF-fiber A_t of A there is a t-definable tuple $\alpha_t \in \Gamma$ such that A_t is α_t-invariant.

Note that covariance makes sense even when the domain or the range of the function f has no VF-coordinates, in which case the contraction $f_\downarrow$ would be the empty function. In particular, if B is a subset of RV^m or even $\mathcal{P}(\mathrm{RV}^m)$ then f being proper covariant simply means that every $f^{-1}(x)$ is proper invariant. Then A being proper invariant may also be construed as the condition that its projection into the RV-coordinates is proper covariant.

Referee R098 [Error] – Fiberwise invariance does not supply global bounds.
Original source line 2008. The asserted characterization drops the global boundedness conditions in Definition 6.2. For example, let $A = 1^\sharp \times \mathrm{RV}$ and let $f : A \longrightarrow \mathrm{RV}$ be the second projection. Every fiber $f^{-1}(t)$ is proper invariant (over t), but $A_{\mathrm{RV}} = \mathrm{RV}$ is not doubly bounded, so f is not proper covariant. Retain the boundedness requirements explicitly, or formulate this as a characterization under those requirements. Also, when the target has no VF-coordinates the relevant contraction is a map to the singleton empty tuple; it is not the empty function unless its domain is empty. The extension to powerset-valued functions should specify separately which boundedness condition is imposed on their RV-valued relation.

Example 6.3. A proper invariant set is not required to be doubly bounded. However, if $\boldsymbol{U} \in \mathrm{RV}^{\mathfrak{db}}[*]$ then $\mathbb{L}\boldsymbol{U}$ is indeed a doubly bounded proper invariant set.

Remark 6.4. In defining proper covariance above, if A is closed then there is no need to demand that $f(A)_{\mathrm{RV}}$ be doubly bounded, because it is implied by the other conditions, as follows. There is an $\mathfrak{o}$-partition $p : A \longrightarrow \Gamma$ such that, for each open polydisc $\mathfrak{b}$ in question, that is, for each open polydisc $\mathfrak{b} \subseteq A$ of the form $\mathfrak{o}(a, p(a))$, $f(\mathfrak{b})$ lies in the range of some VF-fiber of f. By Lemma 2.53,